

Generation Interconnection System Impact Study Report

For Queue Project Q24-2413

Cypress Creek 100 kV
October 23, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2413. The Interconnection Customer proposes a 5.0 MW solar photovoltaic facility interconnecting at the Cypress Creek 100 kV substation in the Indiana area. The request was received on May 30, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 389 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,128,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Laotto Substation / Allings Crossing 100.0 kV is overloaded to 100.9% of its 184 MVA emergency rating for the loss of the Jones Chapel / Dekalb 500.0 kV. The project contributes 13.4% DFAX (0.7 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Allings Crossing / Lotto 100.0 kV	Reconductor line	\$1,128,000	100% (\$1,128,000)
	Total allocated to project		\$1,128,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.